

The Galactic Tidal-Radius Vignette: Recovering the Jacobi Law and its Anisotropic Invariant

A worked example distributed with NestyNet-SR

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Abstract

This note documents the galactic tidal-radius theory-discovery study distributed with NestyNet-SR as `examples/jacobi_tidal`. From mock star-cluster tables the pipeline discovers the anisotropic tidal invariant $4\Omega^2 - \kappa^2$ through its learned general-affine determining operator and recovers the circular-orbit Jacobi radius law in closed form, unifying the classical Hill/Roche and flat-rotation-curve limits. The study also measures the gradient-noise envelope of the symmetry certificate and exhibits principled abstention at 1% observational noise. It serves as the controlled synthetic companion to the SPARC baryonic-acceleration vignette of Paper III NestyNet2026b.

1 The problem

As a theory-discovery case study with mock astronomical observables, we ask the pipeline to recover the circular-orbit Jacobi (tidal) radius of a star cluster or satellite in an axisymmetric galactic host,

$$r_J = \left(\frac{\mu}{4\Omega^2 - \kappa^2} \right)^{1/3}, \quad \mu = GM_{\text{cl}}, \quad (1)$$

where $\Omega = v_c/R$ is the circular frequency and κ the epicyclic frequency of the cluster's orbit. The tidal denominator $q = (2\Omega)^2 - \kappa^2$ is an *anisotropic* indefinite quadratic form, and two textbook approximations are its named limits. For a Keplerian rotation curve ($\kappa = \Omega$) it reduces to the Hill/Roche radius $r_J = R(m/3M)^{1/3}$ (with m the satellite mass and M the host mass enclosed within the orbital radius R), and for a flat rotation curve ($\kappa^2 = 2\Omega^2$) to the classic $r_J = R(m/2M)^{1/3}$ tidal radius (the King limit). Discovering q from data requires the generator $V = \kappa \partial_\Omega + 4\Omega \partial_\kappa$. The isotropic boost generator $\kappa \partial_\Omega + \Omega \partial_\kappa$ of the fixed witness menu maps q to $6\Omega\kappa$ rather than to zero, so the vignette genuinely exercises the learned general-affine determining operator of the generalized-symmetry layer NestyNet2026b and not a named detector.

2 Table design and its two lessons

Each table row is one cluster in its local tidal environment. We draw six host families with power-law rotation curves of logarithmic slope $\alpha \in \{-\frac{1}{2}, -\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$ (spanning the Keplerian through rising-curve regimes), galactocentric radii $R \in [2, 50]$ kpc, cluster masses $M_{\text{cl}} \in 10^{3.5-7} M_\odot$, and

per-row scatter in the local slope, with $\kappa^2 = 2(1+\alpha)\Omega^2$ and a 1% lognormal observational error on r_J . The regression inputs are (μ, Ω, κ) with dimension vectors $[\mu] = L^3 T^{-2}$ and $[\Omega] = [\kappa] = T^{-1}$, and no other information is supplied.

Two design facts matter and are themselves physically instructive. First, a *single* power-law host is uninformative. On one host κ/Ω is constant, every row lies on one ray of the (Ω, κ) plane, and the determining operator correctly reports a large degenerate symmetry algebra (including the isotropic boost) rather than the tidal invariant. The per-host alias $r_J^3 \propto \mu/\Omega^2$ fits perfectly. At least two distinct slopes pin the quadratic uniquely, and the slope scatter within hosts gives the table two-dimensional support. Second, that support controls *derivative* quality. On a table confined to rays the surrogate fits the function values to 10^{-4} , but the gradient transverse to a ray is then an extrapolation beyond the sampled locus and is in error by 20%. Filling the wedge between the rays samples that direction as well and returns the transverse gradients to interpolation accuracy.

3 Symmetry discovery and its noise envelope

From the trained surrogate (values fitted to 3×10^{-5} , analytic gradients accurate to 7×10^{-5} relative), the affine determining operator recovers the off-diagonal generator pair (0.25, 1.00) (that is, $V = \kappa \partial_\Omega + 4 \Omega \partial_\kappa$) at confidence 0.991, giving the discovered invariant

$$\hat{q} = 4.027 \Omega^2 - \kappa^2, \quad (2)$$

whose correlation with the observed tidal curvature μ/r_J^3 is 0.999999 over more than four decades of tidal curvature. As a by-product the same operator discovers that the tidal curvature is homogeneous of degree 2.000 in the frequencies (Euler output action $\beta = 2$). The gradient-quality envelope is sharp and was measured directly by perturbing exact gradients. The invariant is accepted up to relative gradient errors of $\sim 10^{-3}$, still correctly classified at 3×10^{-3} , and lost by 10^{-2} . At the 1% observational noise of the fiducial table, capacity-controlled surrogate gradients plateau at 1.3×10^{-2} , and the one-shot operator then declines to certify any invariant and does not produce a false one. Only the (along-ray) homogeneity survives, essentially unimpaired. Symmetry discovery from a fitted surrogate thus inherits the surrogate’s derivative quality, the central theme of Paper I NestyNet2026a, and precise data, or curvature-supervised surrogate training, is what buys the full invariant.

4 End-to-end closed-form recovery

The end-to-end pipeline, run with units enabled, with the generalized-symmetry Stage-A layer active, and with 6 000 training and 2 000 validation rows, completes the recovery in closed form. On precise data, Stage A autonomously factors the law along its homogeneity structure, $r_J = f(\mu) \cdot F(\Omega/\kappa) \cdot g(\Omega)$, discovering the dimensionless ratio Ω/κ as an internal coordinate. Stage B then closes the mass factor as a free-exponent power law with exponent 0.3333332 (the cube root of Eq. (1), recovered to five parts in 10^7) and closes the dimensionless profile exactly as $(4-t^2)^{-1/3}$, the cube root of a polynomial in the discovered ratio. The fitted quadratic is $4.000 \Omega^2 - \kappa^2$ to machine precision (the linear coefficient is driven below 10^{-15} and pruned), reached at validation loss 8×10^{-33} with three parameters, so the assembled expression is Eq. (1) itself, with no residual neural factor.

Reaching it depends on the two design facts made explicit above. The wedge-filling ensemble supplies the two-dimensional support without which a reciprocal-cubic or rational approximant is numerically indistinguishable from the exact cube root on any single ratio-restricted subsample, and the closure library must contain a cube-root-of-polynomial map alongside its square-root and reciprocal siblings. With both present, the search rejects several hundred plausible-but-wrong closed forms against a 2.7×10^{-10} neural baseline, including the rational approximants that are only good enough on a restricted subsample, and selects the exact cube root, which is simultaneously the most accurate closure and, at three parameters, the parsimony winner. On the 1%-noise table the same pipeline keeps the neural fit and proposes no closure at all. The noise-limited surrogate cannot be certified below the acceptability gate (validation loss 5.9×10^{-6} against a 4.1×10^{-6} threshold), so the search declines to enter the closure stage rather than fit a spurious analytic form to noise. Its nearest structural attempt is instructive, the compound-variable detector reaching for a single tidal factor $z = 2\Omega - \kappa$, one root of $4\Omega^2 - \kappa^2 = (2\Omega - \kappa)(2\Omega + \kappa)$, which it rejects because a lone linear factor leaves a systematic residual. As with the symmetry layer, the closure inherits the surrogate’s precision. The exact law is recovered on precise data, while noise is met with abstention rather than a false positive.

Figure 1 summarizes the vignette. Panel (d) is the theory-level payoff. The two classic approximations fail away from their home regimes exactly as they should, while the single discovered law (one invariant, one cube root) explains every tidal environment at the observational floor. The vignette thus exhibits the intended division of labor. The generalized-symmetry layer discovers the anisotropic carrier that no fixed library contains, the separability machinery factors the law along the symmetry’s homogeneity structure, and the closure rules recover both the rational exponents and the cube-root profile in closed form.

5 Reproduction

The example directory contains the cluster tables (`data/jacobi_master.csv` and the derived `jacobi_v1_full.csv`, `jacobi_v2_cube.csv`, `jacobi_v3_carrier.csv`, with provenance in `data/jacobi_manifest`), the run logs and reports under `results/`, and this figure under `figures/`. The end-to-end invocation is recorded in `results/reroll_harness.log` and follows the fixed benchmark configuration of Paper III with the generalized-symmetry flags enabled:

```
python3 -u -m nestynet_sr.run_SR \
  --filepath examples/jacobi_tidal/data/jacobi_v1_full.csv \
  --units_basis "L,T,M" --y_units "[1,0,0]" \
  --x_units "[[3,-2,0],[0,-1,0],[0,-1,0]]" \
  --gs-stagea --gs-general-affine --gs-noise-calibrated-promotion \
  --ndata_train 6000 --ndata_val 2000
```

References

- [Ibata et al.(2026a)] Ibata, R. A., Tenachi, W., Diakogiannis, F., Ibata, N., & Shankar, A. 2026a, NestyNet. I. Physics Functions Are Hard to Fit with Neural Networks: A Framework for Accurate Surrogates and Analytic Derivatives, ApJ, in preparation
- [Ibata et al.(2026b)] Ibata, R. A., Tenachi, W., Diakogiannis, F., Ibata, N., & Shankar, A. 2026b, NestyNet. III. Symbolic Regression from Analytic Neural Surrogates, ApJ, in preparation

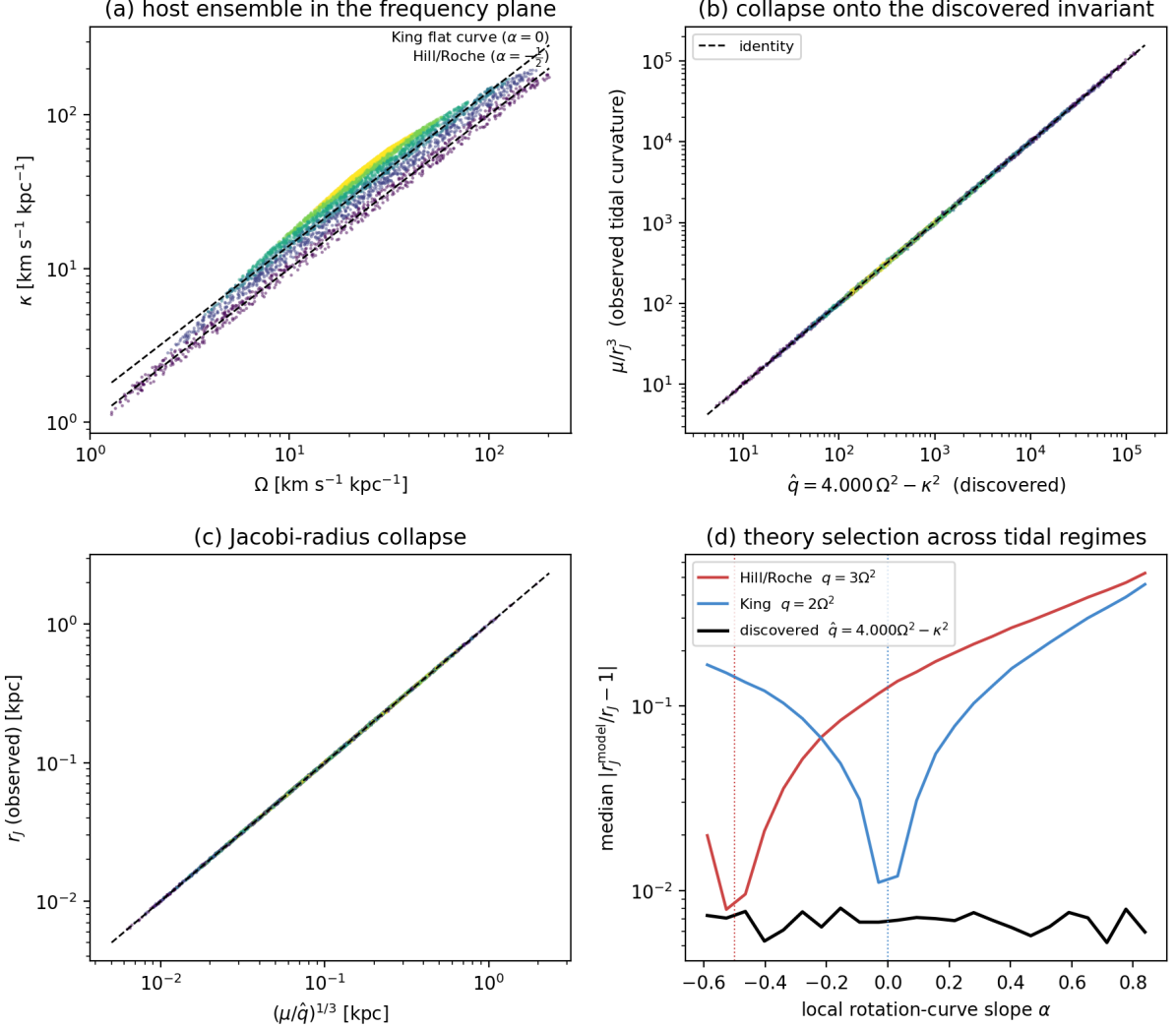


Figure 1: The tidal-radius vignette. (a) The six-host ensemble in the (Ω, κ) plane. The dashed rays are the Keplerian (Hill/Roche) and flat-curve (King) limits, and the per-row slope scatter fills the wedge between host families. (b) Collapse of the observed tidal curvature μ/r_J^3 onto the discovered invariant $\hat{q} = 4.000\Omega^2 - \kappa^2$, over more than four decades. (c) The assembled Jacobi-radius law against the observations. (d) Theory selection across tidal regimes. The Hill/Roche and flat-curve approximations are exact only at their home slopes ($\alpha = -\frac{1}{2}$ and $\alpha = 0$, dotted lines), while the discovered law, having recovered the invariant exactly, sits flat below 1%, at the observational noise floor, across the full range.